

6 (a) DCPIP is an indicator and can be used to determine the rate of respiration of organisms such as yeast.

(i) State the category of indicators to which DCPIP belongs.

..... [1]

(ii) Describe the colour change that occurs in DCPIP during experiments to determine the rate of respiration.

..... [1]

(b) An investigation was carried out to determine the effect of temperature on the rate of respiration of yeast.

- A suspension of yeast cells was added to a test-tube containing glucose solution.
- A further four test-tubes were set up in the same way.
- One test-tube was placed in a water-bath at 10 °C for 5 minutes.
- DCPIP was added to the test-tube and the time taken for the DCPIP to change colour was measured.
- The experiment was repeated using the other test-tubes at 20 °C, 30 °C, 40 °C and 50 °C.

The results are shown in Fig. 6.1.

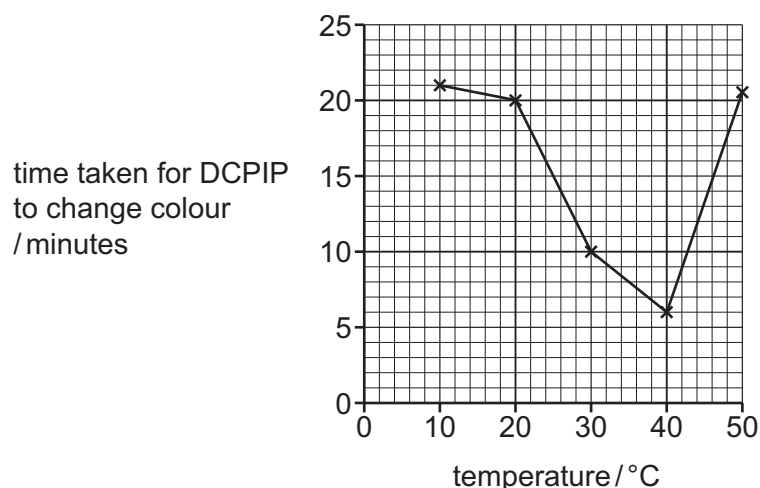


Fig. 6.1

(i) State **two** variables that need to be kept constant in this experiment.

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..... [2]

(ii) Explain the results shown in Fig. 6.1.

[3]

[Total: 7]